

Human Respiration and Heart Rate Estimation Using Wi-Fi Channel State Information and Machine Learning

Estiak Ahamed Sazid, Asaduzzaman, and Mahfuzulhoq Chowdhury

Dept of Computer Science and Engineering, Chittagong University of Engineering and Technology, Bangladesh

Email: u1804051@student.cuet.ac.bd, asad@cuet.ac.bd, mahfuz_csecuet@yahoo.com

Abstract—An automated respiration and heart rate estimation system through signal change analysis would be helpful for early disease prediction. Traditional methods frequently require attached sensors, rendering them unsuitable for continuous monitoring in home environments. The existing channel state (CSI) information based works did not investigate both heart and respiration rate prediction using both machine (ML) and deep learning (DL) techniques with higher accuracy. To outperform these issues, this paper takes advantage of the ubiquitous nature of Wi-Fi signals to estimate human respiration (RR) and heart rate (HR) using Wi-Fi CSI data. The proposed system captures CSI data using standard Wi-Fi hardware, demonstrating how the human body influences electromagnetic signals as they travel through space. A new feature extraction technique that combines shapelet transform and Fast Fourier Transform (FFT) is introduced to improve signal representation and allow for accurate estimation of vital signs. Several ML and DL learning models were implemented and rigorously evaluated. The simulation results hinted that the Extra Trees model shows higher R-squared and lower MAE than other ML and DL models. The comparison results showed that the proposed scheme has at least 5% higher R-Squared and 24% lower MAE value than previous works

Keywords—Channel state information, heart rate, Respiration, Wi-Fi, Machine Learning.

I. INTRODUCTION

Human vital sign estimation that includes both heart and respiration rate analysis are critical in monitoring and understanding an individual's physical condition. To meet the urgent demand for low-cost, non-intrusive biometric monitoring, several studies have been published that considered Wi-Fi CSI data (i.e., channel state information) for people's health activity monitoring [1]. CSI can investigate how the body affects EM signals over not only time domain but also both frequency domain and space domain. Waveform analysis of CSI allows for the identification of heartbeat, respiration, and other physiological processes [2], [3], and [4]. Traditional methods of vital sign tracking, such as capnography [5] and electrocardiogram (ECG) [6], frequently employ the use of attached sensors. However, these dedicated devices have limitations that make them unsuitable for home settings. The availability and ubiquitous nature of Wi-Fi devices has sparked a significant interest in Wi-Fi-based breathing and heartbeat detection [7], [8]. Among the existing works, the authors of [9] used FMCW technology (frequency modulation) to detect people actions using mmWave radar. However, the high cost of FMCW technology makes it

inaccessible to many healthcare facilities. RFID technology is a viable alternative for contactless patient monitoring [10]. While RFID is an intriguing approach, it is based on the application of RFID tags to patients, which may be perceived as invasive. In light of these limitations, researchers have investigated the potential of Wi-Fi radio signals for tracking human vital signs [11]. Unlike RSSI metric, CSI data can provide detailed information about the signals amplitude value and phase value across multiple subcarriers in a Wi-Fi channel. The Intel 5300 model Wi-Fi cards [12] and Atheros CSI Tool [13] are currently used to extract compressed CSIs from compatible networks. Machine (ML) and deep (DL) learning techniques could automatically process patterns and features from CSI data and classify vital signs [14], [15]. Traditional methods for estimating heart rate and respiration rate include contact based photoplethysmography [16], respiratory inductance plethysmography (RIP), and contactless techniques such as Wi-Fi-based sensing, thermal imaging, and optical methods [17]. Contact-based techniques require the patient to wear a device, which can be irritating at times, whereas traditional contact-less techniques are not much cost-effective [18]. By offering hassle-free treatment and saving doctors time and effort when it comes to patient monitoring, non-contact approaches can enhance context settings. Since noninvasive health monitoring techniques don't require any instruments that physically enter the body or break the skin, they are advantageous for long-term use, patient comfortness, and regular monitoring. Several Wi-Fi CSI-based studies on vital sign estimation are currently available in the literary works ([19], [20], [21], [22], [23], and [24]). The existing works did not investigate both heart (HR) and respiration rate (RR) determination simultaneously using ML techniques and non-invasive health monitoring technologies. Most of the studies did not collect real-time data and did not look into rigorous preprocessing and feature extraction procedures. The majority of works have low accuracy values and high error rates when estimating heart rate or respiratory rate. They did not use ML and DL techniques simultaneously for non-contact and CSI-based heart and respiration rate estimation. To outperform these limitations, this paper presents a contactless automated system for human respiration (RR) and heart rate (HR) estimation by utilizing Wi-Fi CSI data, IoT, and learning techniques. This paper prepares a large dataset to train and test the developed model. This paper offers a new feature extraction technique that combines the shapelet transform and the Fast Fourier Transform (FFT). This paper uses the shapelet transform to identify the best candidate shapes for accurately matching respiration and heart rate signals. This

paper examines five different ML and DL models to determine the best model for estimating heart rate and respiratory rate from CSI data. Next, Section two offers a literary works discussion. Section III delivers the proposed scheme. Section IV displays the comparison results and Section V captures a summary of the work.

II. LITERATURE REVIEW

In [15], the authors described a method for monitoring a patient's respiration rate while sleeping using Wi-Fi CSI. Three feature extraction techniques (e.g., Relief, mRMR, and Lasso) are used to extract key features from the dataset. The authors employed KNN algorithm that predicts the health status of a user based on the features chosen. Their work lacks high accuracy in respiration rate estimation (i.e., 85%). In [16], the authors described a method for extracting heart and breathing rate from PPG signals using wavelet decomposition and other signal processing techniques. They did not compare their work to other existing works. In [7], the authors used CSI phase difference analysis to estimate vital signs. They used DTW processing to select the most sensitive subcarrier for vital sign detection, which was then analyzed using FFT. They didn't use any ML or DL algorithms for vital sign classification. Their works also had a high mean absolute error value (MAE) for heart rate estimation (i.e., 4.82). The study in [19] described a CSI-based breathing rate and heart rate estimation method using Wi-Fi technologies. Their method only worked for single and double individuals sharing a bed. They did not use any ML or DL techniques for vital sign classification. They suffered from issues like dynamic movements and high implementation costs. In [20], the author used Principal Component Analysis (PCA) and DFT technology to extract respiratory rates from signal components. They did not explain any learning techniques for heart rate classification. In [22], the authors discussed respiration monitoring techniques based on patterns and models. They investigated the use of Fresnel reflection and diffraction models with Wi-Fi signals to detect human respiration and suffered from high implementation cost issue. In [23], the author used an FFT and DWT mechanism to extract features from CSI data. They only achieved 65% accuracy in heart rate classification.

III. PROPOSED SCHEME

Our proposed CSI-based vital sign (RR and HR) estimation system is depicted in Figure 1, which comprises several phases: using Atheros Wi-Fi chips to collect wireless signals and CSI data, pre-processing and labeling the CSI data, using feature extraction techniques to find the best features, training learning models (ML and DL), comparing and selecting the best prediction model for respiration and heart rate estimation.

A. CSI Data Collection

To collect CSI raw data, we used two desktop computers equipped with Atheros Wi-Fi AR5BMD222 NIC cards. One of the desktop computers served as a transmitter and another as a receiver. Two desktop computers were placed 2.5 meters apart in the room. A chair was strategically placed in the middle of the two desktops. During data collection, two individuals were in charge of sending and receiving commands on the desktop computers. One person acted as the subject during the CSI data collection process. We have collected 34 data sets (24 for training and 9 for testing) and nearly 90000 packets of data. Figure 2(a) gives a demo

regarding the simulation setup, while Figure 2(b) depicts the CSI data collection process.

B. Data Preprocessing and Labeling

The pre-processing phase consists of extracting amplitudes from raw CSI data, selecting subcarriers, reducing noise, and filtering. We started the preprocessing phase by extracting amplitude information from the raw CSI data. Apparently, after amplitude extraction, the CSI data is converted into a CSV file from the dat file (see Figure 3). Following that, we used sub-carrier selection to identify the most important sub-carriers that help capture the subject's respiratory and heart signals. Wi-Fi signals are sent via multiple subcarriers. Not all subcarriers contain useful data for respiration and heart rate monitoring. The top 180 subcarriers with the highest variances from 224 are chosen for further analysis (see Figure 4). To improve the quality of the extracted signals, we used noise reduction and filtering techniques, specifically the Hampel filter (outlier removal) and Running Mean Filters (signal smoothing), to reduce noise and ensure cleaner signal representation (see Figure 5). We then converted the preprocessed data into frequency-based segmentation to distinguish between respiration and heartbeat signals (Figure 6). Heart rates typically range from 60 to 90 beats per minute, with corresponding frequency ranges of 1 Hz to 1.5 Hz. Respiration rates typically range from 12 to 30 breaths per minute, with corresponding frequencies ranging from 0.2 to 0.5 Hz. To isolate these specific frequency ranges, we applied a band pass filter. This segmentation step is critical because it separates the slower respiratory movements from the faster heartbeat movements, allowing for more precise feature extraction. Following frequency-based segmentation, we proceeded with data labeling to ensure an accurate correlation between measured signals and physiological ground truth values. Each sample consisted of 15 seconds of continuous data, which was sufficient to capture meaningful physiological variations. Each sample was labeled with the corresponding ground truth value. We labeled respiration rates based on the subject's selfcounting of breaths. To obtain accurate heart rate measurements, we used a pulse oximeter device.

C. Feature Extraction

After labeling the data, we completed the feature extraction task. The shapelet transform (ST) can identify the most informative subsequences (shapelets) from the time series (TS) data that can differentiate between different classes or patterns [25]. The key idea is to find shapelets that are maximally representative of one class while being minimally present in others. We obtained seven distinct shapelets for the respiration data after transforming them. These shapelets capture the key patterns in the respiration signal, as shown in Figure 7. Similarly, we extracted eight different shapelets from the heart rate data using the shapelet transform process. These shapelets encapsulate the distinct patterns found in the heart rate signal, as illustrated in Figure 8. After shapelet transforming our data, we used the fast fourier transform (FFT). We extracted two features using FFT. They are the frequency components with the largest magnitude along with standard deviation. Apparently, combining these two features with previously extracted features yields final feature matrices. The transformed respiration and heart rate data, represented by shapelets and frequency domain features, were then used to create feature matrices for a variety of ML and DL classification tasks.

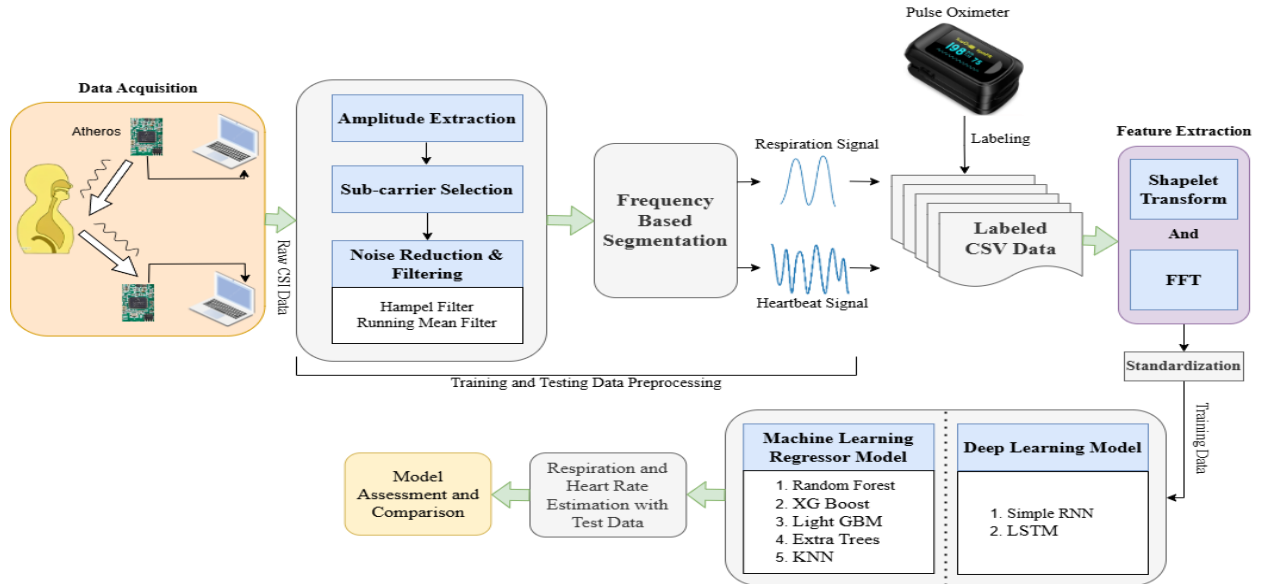


Fig. 1. Methodology diagram of our proposed system

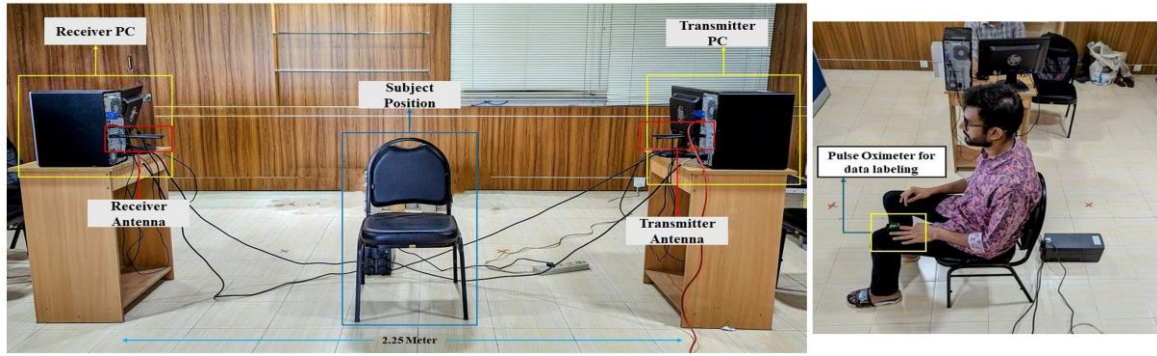


Fig. 2. (a) Internal set up of the experiment room (b) collecting CSI data from a participant

Sub 0	Sub 0	Sub 0	Sub 0	Sub 1	Sub 1	Sub 1	Sub 1	Sub 2	Sub 2	...	Sub 53	Sub 53	Sub 54	Sub 54	Sub 54	Sub 54	Sub 54
RXTX 0/0	RXTX 0/1	RXTX 1/0	RXTX 1/1	RXTX 0/0	RXTX 0/1	RXTX 1/0	RXTX 1/1	RXTX 0/0	RXTX 0/1	...	RXTX 1/0	RXTX 1/1	RXTX 0/0	RXTX 0/1	RXTX 1/0	RXTX 1/1	RXTX 0/0
0	28.876173	43.841921	41.139434	41.975562	30.663259	43.726359	40.875330	42.529743	31.658376	43.588862	...	28.457180	39.093955	34.360035	28.603380	26.190933	38.925398
1	34.222615	43.751513	46.768856	41.702617	35.871495	43.577253	46.636821	42.321063	36.824159	43.545886	...	34.583356	38.585372	40.539232	27.752463	32.214142	38.773713
2	33.408405	41.796954	46.690564	39.647309	35.314789	41.620264	46.536369	40.685198	36.418705	41.490035	...	34.708513	36.885978	40.026411	27.395723	32.095150	36.867256
3	34.295908	42.108266	46.743191	40.541916	35.594278	41.923444	46.638893	41.253186	37.004441	41.892095	...	34.527062	37.027751	40.539232	26.928469	31.535100	37.528932
4	34.608978	44.884661	47.617024	42.831429	36.359861	44.823161	47.441130	43.535316	37.516639	44.547382	...	35.650209	39.782718	41.194209	30.107239	32.263421	39.674076

5 rows x 224 columns

Fig. 3. Glimpse of sample data after amplitude extraction

TABLE I
COMPARISON AMONG ML AND DL MODEL FOR RESPIRATION RATE (RR) ESTIMATION

Model	R squared	MAE	MSE	RMSE	MAPE
RF	.9561	.2325	.467	.6834	1.1332
XGB	.9367	.4471	.6728	.8202	2.217
LGBM	.9171	.5281	.8798	.938	2.7114
ETs	.9662	.2069	.3595	.5995	.9963
KNN	.9454	.2308	.5803	.7618	1.1219
RNN	.937	.4389	.674	.821	2.0806
LSTM	.9636	.2706	.389	.6237	1.2688

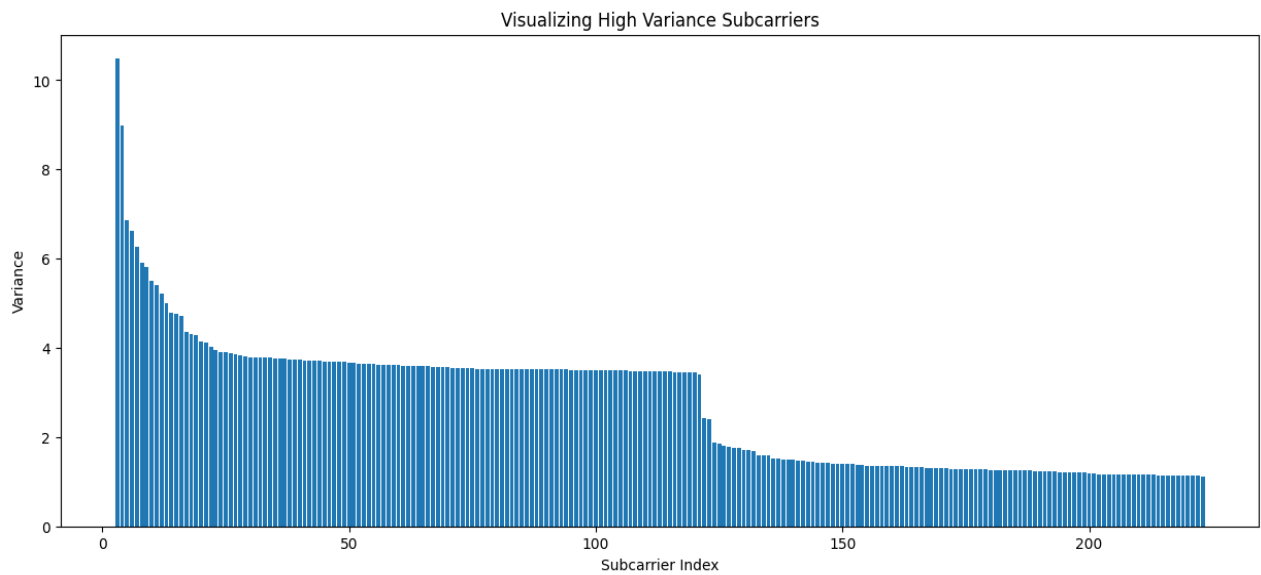


Fig. 4. Significant subcarrier based on higher variance

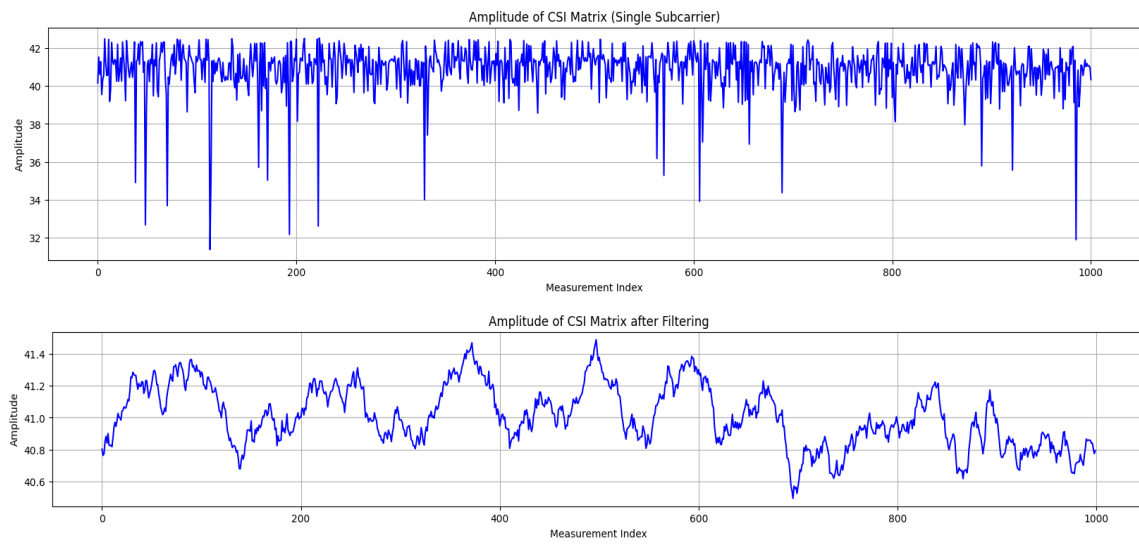


Fig. 5. Noise reduction and filtering results

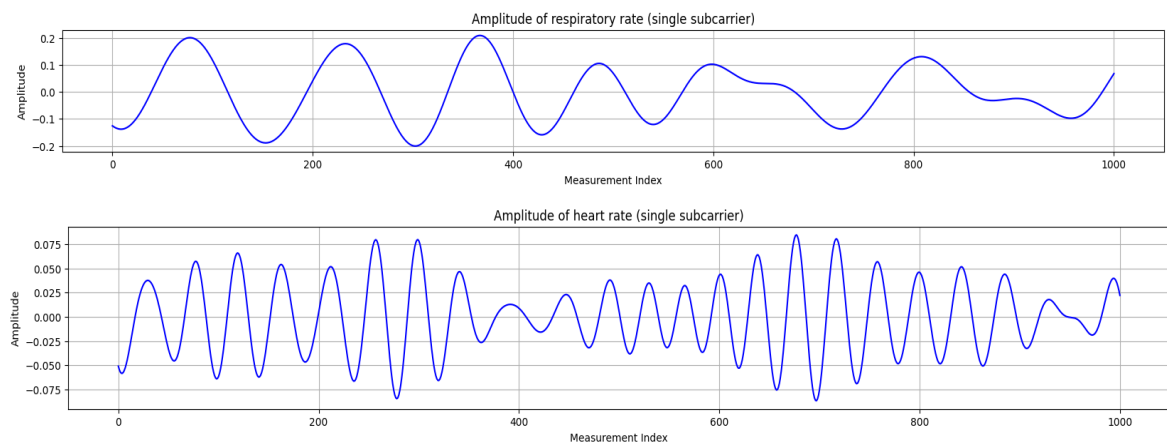


Fig. 6. Frequency based respiration and heart rate signal segmentation

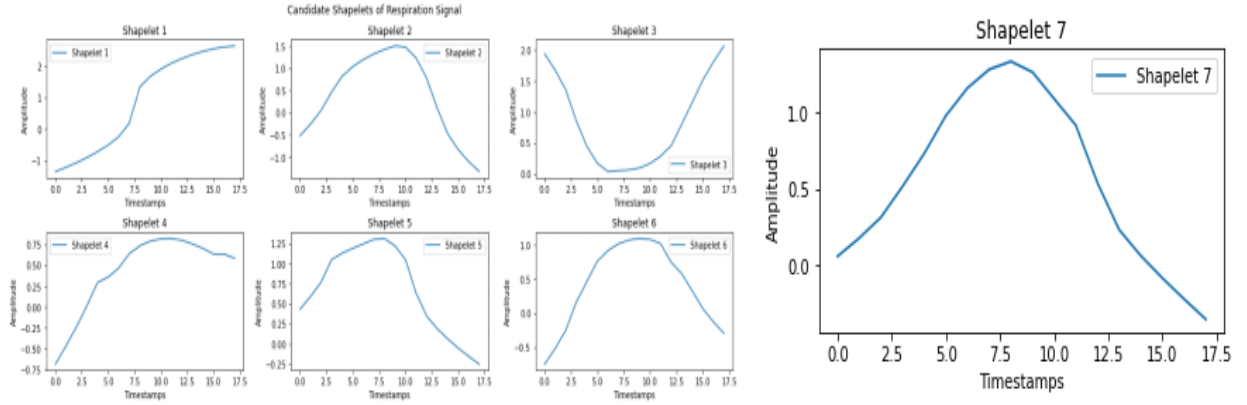


Fig. 7. Shapelets extracted from the respiration CSI data

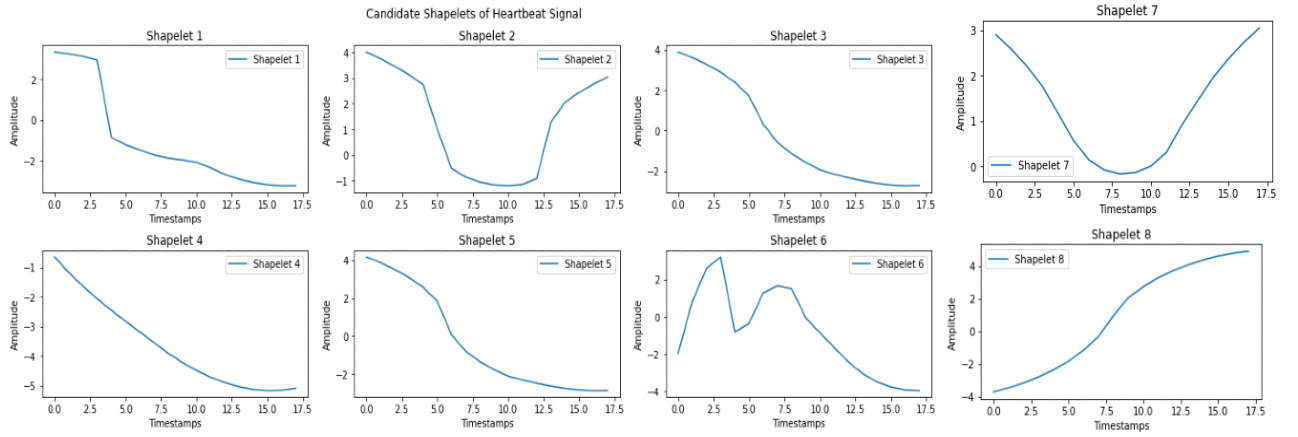


Fig. 8. Shapelets extracted from the heart rate CSI data

TABLE II
COMPARISON AMONG ML AND DL MODEL FOR HEART RATE (HR) ESTIMATION

Model	R squared	MAE	MSE	RMSE	MAPE
RF	.7902	1.0639	5.9667	2.4427	1.4834
XGB	.7541	1.4363	6.9302	2.6325	2.0029
LGBM	.7155	1.7096	8.0196	2.8319	2.3857
ETs	.8006	1.0339	5.6546	2.3779	1.4468
KNN	.7765	1.0376	6.3009	2.5102	1.4497
RNN	.7791	1.4945	6.1538	2.4807	2.0458
LSTM	.8003	1.1808	5.253	2.2919	1.6049

IV. RESULTS AND ANALYSIS

We used five different machine and deep models for our respiration and heart rate estimation task to ensure reliable and accurate results. These models comprised the random forest regressor (RF), XGBoost (XGB), LightGBM (LGBM), extra trees regressor (ETs), and K-Nearest Neighbors (KNN) regressor, Recurrent Neural Network (RNN), and LSTM. We have selected the best prediction model (from all ML and DL models) for heart (HR) and respiration (RR) rate estimation

by examining R squared value (accuracy), MAE (mean absolute error), MSE, RMSE, and MAPE error metrics. Tables I and II show that for both respiration (RR) and heart rate (HR) prediction, the Extra Trees (ETs) regressor outperformed other regression models (RF, XGB, LGBM, KNN) and DL models in terms of higher r-squared value and lower error values (e.g., MAE, MAPE, RMSE, MSE value). The extra tree regressor (ETs) produces the best results for respiration rate estimation, with values of .966 r-squared, .20

MAE .35 MSE, .59 RMSE, and .99 MAPE. For heart rate estimation, the extra tree regressor (ETs) achieves .8006 r-squared, 1.03 MAE, 5.65 MSE, 2.37 RMSE, and 1.44 MAPE value. Extra Trees' superior performance could be attributed to its inherent ability to handle high-dimensional data effectively while reducing over fitting via random feature selection and multiple tree averaging. The shapelet transform-based feature extraction likely improved the meaningful patterns in the data. Table III illustrates a detailed comparison between a recent previous work [21] and our proposed work. The results also revealed that the proposed scheme achieves at least 5% higher r squared value and 24 percent lower MAE values than existing works.

TABLE III
COMPARISON WITH EXISTING WORKS

Points	Existing [21]	Our proposed work
Tools used	Intel 5300 NIC	Atheros CSI tool
Filtering	Hampel filter	Hampel and running mean filter
Feature extraction	FFT	FFT and shapelet transform
R squared for RR	.9159	.9662
MAE for RR	.3914	.2069
R squared for HR	.7593	.8006
MAE for HR	1.3768	1.0339

V. CONCLUSION

This paper presents a comprehensive framework for estimating human respiration (RR) and heart rate (HR) using Wi-Fi CSI data, machine learning, and Internet of Things technologies. The framework includes data acquisition, advanced signal pre-processing, and frequency-based segmentation to accurately capture respiration and heart rate signals. A novel feature extraction technique combining shapelet transform and FFT was used to improve the representation of these physiological signals. Several machine (ML) and deep learning (DL) models were implemented and thoroughly tested for heart rate (HR) and respiration (RR) rate estimation. The Extra Trees regressor model outperformed the other models, with an r-squared value of .9662 for respiration (RR) and .8006 for heart (HR) rate estimation. In the future, advanced feature extraction and spatial filtering techniques can be used along with this work to improve the estimation accuracy of heart (HR) and respiration rates (RR) in a more noisy and multiperson setting.

REFERENCES

[1] Y. Ma et al., "Wifi sensing with channel state information: A survey," *ACM Computing Surveys*, vol. 52, no. 3, pp. 1–36, 2019.

[2] S. Lee et al., "Design and implementation of monitoring system for breathing and heart rate pattern using WiFi signals," *15th CCNC, USA*, 2018, pp. 1–7.

[3] H. Li et al., "Wi-Motion: A Robust Human Activity Recognition Using WiFi Signals," *IEEE Access*, vol. 7, pp. 153287–153299, 2019.

[4] H. Wang et al., "RT-Fall: A Real-Time and Contactless Fall Detection System with Commodity WiFi Devices," *IEEE Transactions on Mobile Computing*, vol. 16, no. 2, pp. 511–526, 2017.

[5] J. D'Mello et al., "Capnography," *Indian journal of anaesthesia*, vol. 46, no. 4, pp. 269–278, 2002.

[6] R. M. Laukkanen et al., "Heart rate monitors: State of the art," *Journal of sports sciences*, vol. 16, no. 1, pp. 3–7, 1998.

[7] X. Wang et al., "On csi-based vital sign monitoring using commodity wifi," *ACM Transactions on Computing for Healthcare*, vol. 1, no. 3, pp. 1–27, 2020.

[8] B. Tan et al., "Exploiting WiFi Channel State Information for Residential Healthcare Informatics," in *IEEE Communications Magazine*, vol. 56, no. 5, pp. 130–137, 2018.

[9] M. Alizadeh et al., "Remote Monitoring of Human Vital Signs Using mm-Wave FMCW Radar," in *IEEE Access*, vol. 7, pp. 54958–968, 2019.

[10] X. Liu et al., "Vital Signs Monitoring with RFID: Opportunities and Challenges," *IEEE Network*, vol. 33, no. 4, pp. 126–132, 2019.

[11] Y. Ge et al., "Contactless WiFi Sensing and Monitoring for Future Healthcare - Emerging Trends, Challenges, and Opportunities," *IEEE Reviews in Biomedical Engineering*, vol. 16, pp. 171–191, 2023.

[12] Z. Wang et al., "A survey of user authentication based on channel state information," *Wireless Communications and Mobile Computing*, vol. 2021, pp. 1–16, 2021.

[13] S. Tan et al., "Commodity WiFi Sensing in Ten Years: Status, Challenges, and Opportunities," *IEEE IoT Journal*, vol. 9, no. 18, pp. 17832–17843, 2022.

[14] S. M. Hernandez et al., "WiFi Sensing on the Edge: Signal Processing Techniques and Challenges for Real-World Systems," *IEEE Communications Surveys and Tutorials*, vol. 25, no. 1, pp. 46–76, 2023.

[15] M. I. Khan et al., "Tracking vital signs of a patient using channel state information and machine learning for a smart healthcare system," *Neural Computing and Applications*, pp. 1–15, 2021.

[16] R. Jaafar et al., "Estimation of breathing rate and heart rate from photoplethysmogram," *ICEEI conf, Malaysia*, 2017, pp. 1–4.

[17] C. Xu et al., "Cardiacwave: A mmwave-based scheme of non-contact and high-definition heart activity computing," *ACM IMWUT journal*, vol. 5, no. 3, pp. 1–26, 2021.

[18] J. Liu et al., "Tracking vital signs during sleep leveraging off-the-shelf wifi," *16th MobiHoc*, 2015, pp. 267–276.

[19] J. Liu et al., "Monitoring Vital Signs and Postures During Sleep Using WiFi Signals," *IEEE IoT Journal*, vol. 5, no. 3, pp. 2071–2084, 2018.

[20] T. Kanda et al., "Respiratory Rate Estimation Based on WiFi Frame Capture," *IEEE CCNC, USA*, 2022, pp. 881–884.

[21] X. Zhang et al., "Wital: A COTS WiFi Devices Based Vital Signs Monitoring System Using NLOS Sensing Model," *IEEE Trans. on Human-Machine Systems*, vol. 53, no. 3, pp. 629–641, 2023.

[22] D. Zhang et al., "wifi csi-based vital signs monitoring," in *Contactless Vital Signs Monitoring*, Academic Press, 2022, pp. 231–255.

[23] J. A. Armenta-Garcia et al., "Mining wi-fi channel state information for breathing and heart rate classification," *Pervasive and Mobile Computing*, vol. 91, pp. 1–10, 2023.

[24] M. Atif et al., "Covid-beat: A low-cost breath monitoring approach for people in quarantine during the pandemic," *Journal of Computational Design and Engineering*, vol. 9, no. 3, pp. 992–1006, 2022.

[25] J. Grabocka et al., "Learning time-series shapelets," *20th ACM SIGKDD int. conf. on Knowledge discovery and data mining*, 2014, pp. 392–401.